

CURRICULUM VITAE

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Academic Profile

Applied data science professional with a background in business, supply chain, and military logistics, preparing for Ph.D. study and adjunct teaching in data science, analytics, archaeogenomics, and quantitative methods. Research and teaching interests include machine learning, statistical modeling, data management, bioinformatics, population genetics, phylogenetic analysis, educational assessment, and applied analytics in health and higher education.

Teaching Interests

- Data science and introductory analytics
- Statistics, data management, and research methods
- Machine learning, visualization, and database fundamentals
- Educational assessment, program evaluation, and health/biomedical analytics
- Anthropology, humanities, and interdisciplinary social inquiry, with a focus on cultural analysis, human behavior, and connecting theoretical concepts to real-world contexts

Education

2024-2026	M.S. – Applied Data Science (Biomedical Engineering Focus), University of Florida, Gainesville, Florida
2019-2021	B.A. – Religion & Cultural Studies; Minor, Judaic Studies, University of Central Florida, Orlando, Florida
2017-2018	A.S. – Supply Chain Management, Seminole State College of Florida, Sanford, Florida
2017-2019	A.A. – General Education, Seminole State College of Florida, Sanford, Florida

Selected Graduate Projects

Capstone Project — Machine Learning for Genetic Variant Classification (Spring 2026)

- Built an end-to-end Python and SQL pipeline to ingest, clean, and process biological datasets.
- Developed classification models to distinguish pathogenic vs. benign variants.
- Evaluated model quality with ROC curves, confusion matrix, and F1-score.
- Delivered reproducible workflow artifacts and technical documentation for repeatable analysis.

Machine Learning II Project — Cardiovascular Disease Risk Prediction (Fall 2025)

- Compared logistic regression, random forest, and 1D CNN models on Kaggle cardiovascular datasets.
- Evaluated predictive trade-offs between accuracy, interpretability, and computational efficiency.
- Assessed model performance with ROC-AUC, F1-score, and precision/recall.

Bioinformatics Project — Transformer Fine-Tuning for Regulatory DNA (Fall 2025)

- Designed and implemented a pipeline to classify regulatory DNA elements and score non-coding variant effects.
- Fine-tuned transformer-based language models on ENCODE, Roadmap Epigenomics, and DeepSEA datasets.
- Benchmarked against CNN baselines using AUROC, PR-AUC, runtime, and cross-cell-type generalization.

Machine Learning I Project — Spotify Data Science Project (Spring 2025)

- Conducted exploratory data analysis and statistical profiling on music metadata and engagement signals.
- Built and compared Logistic Regression and Random Forest classifiers.
- Applied hyperparameter tuning with GridSearchCV to optimize model performance.

Global Religious Demographics Classification (Spring 2025)

- Engineered features from multiple demographic datasets related to global religion projections.
- Applied PCA, UMAP, and t-SNE for dimensionality reduction and visualization.
- Trained logistic regression, random forest, and SVM classifiers to analyze and classify global religious demographics.

Technical Skills

Languages: Python, SQL, R

Analysis and Statistics: Exploratory data analysis, hypothesis testing, descriptive/inferential statistics, regression analysis, classification modeling, PCA, UMAP, t-SNE

Data and Machine Learning Tools: Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Jupyter Notebooks, Excel

Visualization and Reporting: Tableau, Power BI, KPI dashboards, trend analysis, operational reporting

Data Engineering: ETL pipelines, data cleaning, data wrangling, data validation, relational databases, SQL query optimization

Workflow: Git, reproducible analysis, documentation, cross-functional collaboration

Professional Experience

Regional Transit System (City of Gainesville) — Gainesville, FL

Parts Specialist | November 2023 – December 2023

- Managed structured inventory datasets that supported maintenance and operations decision-making.
- Maintained high data accuracy across inventory systems to reduce reconciliation discrepancies.
- Partnered with technicians to ensure timely, accurate data logging and operational reporting.

U.S. Army

Equipment Records Parts Sergeant, Fort Bragg, NC | September 2014 – March 2017

- Took ownership of the organization's equipment dispatching process to adhere to safety and procedural guidelines; enhanced operational continuity practices.
- Led six personnel in the shop office and maintained mission-essential information systems.
- Completed tasks and mission requirements while supporting the organization with dependable service and records management.

Warehouse Operator, Fort Bragg, NC | August 2013 – September 2014

- Processed over 380 repair parts worth \$34,000,000 in funds for the organization.
- Coordinated with adjacent units for parts and resources to ensure equipment readiness and reduce downtime.
- Ensured shop stock and bench stock for three separate sections.

Combat Engineer / Operations Assistant / Vehicle Driver, Joint Base Elmendorf-Richardson, AK | July 2009 – May 2013

- Monitored patrols, air movements, personnel, Afghan Border Police missions, and key leader locations for operational visibility.
- Served as a Route Reconnaissance/Clearance Operator to report, monitor, and analyze improvised explosive device activity.
- Supported field training exercises and live-fire readiness exercises in preparation for combat operations in support of Operation Enduring Freedom.

Customer Service & Sales Roles — Bass Pro Shops and Gander Outdoors

- Applied data tracking, inventory controls, and customer service skills in retail operations.

Teaching and Training Experience

- Supervised and trained 10+ personnel on inventory control, records accuracy, and standard operating procedures.
- Explained procedures, checked work products, and supported team members in fast-paced operational settings.
- Used clear written communication and process documentation to help others follow standards consistently.

Military Education and Training

- Basic Combat Training
- Combat Engineer Airborne
- Route Reconnaissance/Clearance Operator (GWOT)
- Structured Self-Development I and II
- Warrior Leader Course
- Automated Logistical Specialist
- Unit Movement Officer Deployment Planning
- Introduction to Management
- Fundamentals of Intermodal Transportation
- GCSS-Army Overview
- Basic Leadership Course
- Combat Engineer Training

Professional Certifications

- Fundamentals of Deep Learning (NVIDIA)
- OSHA 10-Hour Certification
- Forklift Operator Certification
- Certificate, Unit Movement Officer Course (40 hours)
- Certificate, Automated Logistics Specialist
- Certificate, GCSS-Army Overview